REVIEW ARTICLE

Air Pollution and Lung Cancer: A Review by International Association for the Study of Lung Cancer Early Detection and Screening Committee

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ABSTRACT

Introduction: The second leading cause of lung cancer is air pollution. Air pollution and smoking are synergistic. Air pollution can worsen lung cancer survival.

Methods: The Early Detection and Screening Committee of the International Association for the Study of Lung Cancer formed a working group to better understand issues in air pollution and lung cancer. These included identification of air pollutants, their measurement, and proposed mechanisms of carcinogenesis. The burden of disease and the underlying epidemiologic evidence linking air pollution to lung cancer in individuals who never and ever smoked were summarized to quantify the problem, assess risk prediction models, and develop recommended actions.

Results: The number of estimated attributable lung cancer deaths has increased by nearly 30% since 2007 as smoking has decreased and air pollution has increased. In 2013, the International Agency for Research on Cancer classified outdoor air pollution and particulate matter with

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aerodynamic diameter less than 2.5 microns in outdoor air pollution as carcinogenic to humans (International Agency for Research on Cancer group 1) and as a cause of lung cancer. Lung cancer risk models reviewed do not include air pollution. Estimation of cumulative exposure to air pollution exposure is complex which poses major challenges with accurately collecting long-term exposure to ambient air pollution for incorporation into risk prediction models in clinical practice.

Conclusions: Worldwide air pollution levels vary widely, and the exposed populations also differ. Advocacy to lower sources of exposure is important. Health care can lower its environmental footprint, becoming more sustainable and resilient. The International Association for the Study of Lung Cancer community can engage broadly on this topic.

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Keywords: Lung cancer; Air pollution; Lung cancer risk; Climate change

Introduction

Lung cancer is the leading cause of cancer mortality in both men and women with more than 1.8 million deaths a year worldwide.[1] Globally, the five-year net survival for lung cancer is 10% to 20% and is among the lowest of all types of cancer.[2] In 1950, Doll and Hill[3] suggested tobacco smoking and exposure to outdoor pollutants as the two main causes of lung cancer. Until recently, most cancer control measures and research have focused on tobacco smoking. Lung cancer deaths related to tobacco smoking are projected to decrease in the next 25 years as smoking rates continue to decline. Nevertheless, as smoking-related lung cancers decrease, the proportion of lung cancers in people who have never smoked will continue to increase.[4] As a separate entity, lung cancer in individuals who have never smoked is currently the seventh leading cause of cancer deaths in both sexes worldwide.[5] Increasing evidence indicates that air pollution is a major cause of lung cancer, and the number of estimated attributable lung cancer deaths has increased by nearly 30% since 2007 as smoking has decreased and air pollution has increased.[6]

In 2013, the International Agency for Research on Cancer (IARC) classified outdoor air pollution and particulate matter (PM) with aerodynamic diameter less than 2.5 microns ($PM_{2.5}$, approximately 1/30th the width of a human hair) in outdoor air pollution as carcinogenic to humans (IARC group 1) and as a cause of lung cancer.[7–9] The Global Burden of Disease 2019 study estimated that the percent of the global lung cancer deaths attributable to different potentially modifiable risk factors was 62% for smoking, 15% for $PM_{2.5}$ in outdoor air, 5.8% for secondhand smoke, 4% for household air pollution from use of solid fuels for cooking, and 4% for radon, with occupational exposures and possibly dietary factors contributing the remainder.[9] It has been estimated that in the People's Republic of China, approximately 180,000 deaths annually and 25.8% of disability-adjusted life years due to lung cancer are attributable to air pollution.[10] Table 1 highlights the large contribution of fine particulate matter in the development of lung cancer.

Similarly, to the impact of the reduction in smoking rates, reduction of air pollution is expected to result in a substantial reduction of lung cancer deaths. Smoking and air pollution interact in a positive synergistic fashion[11] and efforts at smoking cessation need to continue unabated. In this white paper, we review the current understanding of air pollution's contribution to lung cancer and what the health care community can do to advocate for mitigation of harms on the individual patient level and at the public health and policy levels.

Types of Air Pollutants

Air pollutants can be categorized as PM or gaseous pollutants. PM is the mixture of solid and liquid particles grouped by aerodynamic diameter. Ambient PM consists of diverse chemical compounds that include polycyclic aromatic hydrocarbons (PAHs), n-Nitroso compounds such as nitrosamines, and other related compounds, which differ depending on their sources (e.g., construction, industry, transportation, and wildfires) and that vary in physical size (i.e., aerodynamic properties). These factors largely determine their transport and removal processes in the air. Deposition sites and clearance pathways within the respiratory tract are also affected. The most harmful to respiratory health are those with $PM_{2.5}$.[12] IARC identifies $PM_{2.5}$ in outdoor air pollution as a group 1 carcinogen due to mechanistic evidence, animal studies, and epidemiologic evidence in humans.[13]

Gaseous pollutants that are potential health hazards include, but are not limited to, ground-level ozone (O_3), volatile organic compounds (VOCs), sulfur dioxide, and nitrogen oxides (NO_x), consisting of nitrogen oxide and nitrogen dioxide (NO_2). Ground-level O_3 is a major component of photochemical smog and is mainly formed by chemical reactions of NO_x with VOCs particularly at warm temperatures. VOCs are compounds that have a high vapor pressure and low water solubility and include aldehydes, ketones, aromatics, and other classes. They are emitted by different industrial sources (e.g., manufacture of paints, synthetic plastic materials, and pharmaceuticals) and indoor

Table 1. Global Burden of Disease 2017 Risk Factor Collaborators' Estimates of Global Lung Cancer Mortality Attributable to Selected Risk Factors in 2017

Exposure	Rank	Number in 1000s	95% Uncertainty Interval
Smoking	1	1190	1150-1230
Ambient particulate matter pollution	2	265	183-351
Household air pollution from solid fuels		85	60-113
Occupational exposure to asbestos	3	191	137-247
Occupational exposure to silica		49	22-77
Occupational exposure to diesel engine exhaust		18	16-20
Occupational exposure to arsenic		9	2-16
Occupational exposure to nickel		9	1-23
Occupational exposure to polycyclic aromatic hydrocarbons		5	4-6
Occupational exposure to cadmium		1	1-1
Occupational exposure to chromium		1	1-2
Diet low in fruits	4	185	77-320
Secondhand smoke	5	100	57-149
Residential radon	6	88	50-139

Note: This table has been constructed from data in cited reference 12. This list of selected risk factors excludes elevated fasting blood glucose, which does appear in the original published cited reference. We excluded it from this table because the causal association between elevated fasting blood glucose and lung cancer has not been firmly established.

sources, such as cooking methods. Sulfur dioxide and NO_x are mainly derived from the combustion of fossil fuels, an important example of which is automobile exhaust. Epidemiologic and toxicology studies suggest that exposure to these gaseous pollutants can lead to respiratory symptoms,[14,15] asthma exacerbation,[16,17] increased emergency department visits, and hospital admissions in relation to airway problems.[18,19]

Measurements of Air Pollutants

Measurement or exposure assessment of air pollutants aims to quantify the levels of air pollution people are exposed to over time in addition to the estimated number of people being exposed. Methods for exposure assessment include, but are not limited to, personal measurement, outdoor air quality monitoring, and computational modeling. Personal measurements with wearable samplers that measure the air in the respective individual's breathing zone are considered the "gold standard" for exposure assessment as it can detect pollutant concentrations in individual microenvironments.[20–22] Nevertheless, epidemiologic studies of relationships between air pollution exposure and cancer require long periods of observation and large populations. It is not feasible with currently available approaches to assess long-term pollutant exposure by means of personal monitoring. Accordingly, the most traditional proxy is assigning measured outdoor air pollution level or pollutant concentrations to the study populations. For instance, both the American Cancer Society[23] and the Harvard "Six Cities"[24] studies estimated the exposure of participants on the basis of long-term average concentrations measured at ambient monitoring stations. Nevertheless, monitoring stations are typically built at a limited number of locations, so that outdoor monitoring data usually cannot reflect spatial gradients of air pollutants, especially for those with spatially heterogeneous distribution.[25,26] Recent studies use more sophisticated methods, such as geographic information systems (GIS)–based models and remote sensing–based models, to address this challenge. Land use regression models, using statistical modeling to generate air pollution prediction maps on the basis of measurements, surrounding land use, and other characteristics (e.g., traffic) obtained through GIS can effectively capture small-scale variability of air pollution with a relatively low demand for data input.[27–30] The European Study of Cohorts for Air Pollution Effects have applied this approach in 17 European cohorts and revealed robust associations between long-term PM exposure and the risk of lung cancer.[31]

Remote sensing–based estimates of air pollution have been increasingly applied in epidemiologic studies.[32] For example, van Donkelaar et al.[33] developed a global model of monthly $PM_{2.5}$ concentration at a spatial resolution of approximately 1×1 km. This approach usually combines satellite-retrieved measures (e.g., aerosol optical depth, which is a measure of the scattered light from all aerosols within the total column between the Earth's surface and the satellite) with other information (e.g., land use, climate, and chemical transport model prediction) to fill the spatiotemporal gaps left by ground monitors with high-quality predictions. This provides estimates of concentrations of pollutants of interest (e.g., $PM_{2.5}$) anywhere in the world.[34,35] In general, most current modeling approaches still ignore exposure from various microenvironments as people spend

considerable times away from their home address, such as at work, and therefore, exposure misclassification is inevitable. To address these challenges, concepts of more integrated methodologies combining traditional air quality monitoring networks and GIS and remote sensing technologies with information on personal behavior/time-activity patterns have been proposed[36,37] (e.g., Stochastic Human Exposure and Dose Simulation[38]). These approaches may help to minimize measurement error in exposure assessment in future studies on the relationship between air pollution and cancer. We must also account for a mobile population and global migration. Many individuals move multiple times throughout their lifetime, within their country or around the world. Tracking a residential history for each individual is important to increase accuracy of exposure, outside of personal measurements.

Mechanisms of Carcinogenesis

Polluted air contains at least two types of carcinogens: PAHs and n-nitroso compounds, such as nitrosamines. Research reveals that $PM_{2.5}$ can promote carcinogenesis in the lungs. $PM_{2.5}$ has high PAH concentrations, which can induce oxidative stress and aryl hydrocarbon receptor activation in human bronchial epithelial cells (BEAS-2B).[39] Aryl hydrocarbon receptor activation promotes lung cancer cell invasion and metastasis.[39] One study found that mortality from lung cancer caused by $PM_{2.5}$ is generally much higher in women than in men.[40] In addition to causing local inflammation and oxidative stress, $PM_{2.5}$ penetrates deep into the lungs and terminal bronchioles, and from there into the circulation, causing systemic inflammation. Recent research has found that after long-term exposure to PM_{10} in a human NSCLC line (A549), there was a decrease in mitotic index and in the percentage of cells in the G2/M stage compared with a control group.[39] Long-term exposure to fine particle air pollutants can produce genotoxicity and mutagenicity. It may increase the risk of lung cancer through inflammatory damage, reactive oxygen species production, and oxidative DNA damage. Therefore, especially in people with a high genetic risk, prolonged air pollution may increase the incidence of lung cancer. Huang et al.[41] reported a synergistic effect between high genetic susceptibility (determined by a polygenic risk score on the basis of genome-wide association studies) and higher $PM_{2.5}$ exposures on lung cancer incidence. EGFR mutant-driven lung cancer is more common in individuals who have never smoked or in those with light smoking history, and in 32,957 lung cancer cases in four cohorts, there was a significant association with $PM_{2.5}$.[42] Furthermore, functional mouse models revealed that pollution causes an influx of macrophages into the lung epithelium and interleukin-1 release, resulting in a progenitor-like cell state within EGFR-mutant lung alveolar type II epithelial cells that fuel and promote tumorigenesis, in a process that can be attenuated with anti–interleukin-1.[42]

Epidemiologic Evidence Linking Air Pollution to Lung Cancer in Never- and Ever-Smoking Individuals

The most often evaluated pollutant in epidemiology studies is $PM_{2.5}$. Other exposure indicators include PM_{10}, total suspended particles, black smoke, and other common components of air pollution, such as nitrogen oxides (NO_x), NO_2, SO_2, and O_3. Several studies included indirect indicators of exposure, such as proximity to roads or traffic volume, but these studies are considered to provide a lower quality of evidence. A challenge in interpreting the associations for $PM_{2.5}$ lies in determining whether $PM_{2.5}$ itself acts as a carcinogenic agent or whether it is a surrogate for the outdoor air pollution mixture or other components of it. One hypothesis is that the existence of gas-phase carcinogens, highly correlated with $PM_{2.5}$ concentration, is the actual cause of lung cancer. Nevertheless, the fact that associations with $PM_{2.5}$ level have been observed in multiple locations with different pollution mixtures is an argument for an effect of $PM_{2.5}$ itself.

Epidemiologic studies have revealed that people living in highly polluted areas are more likely to die of lung cancer than those who do not. This association of exposure to outdoor air pollution with lung cancer risk has been evaluated in cohort and case-control studies conducted in Europe, North America, Asia, and Oceania. The most informative investigations are large-scale prospective cohort studies with individual-level information on lung cancer risk factors, including tobacco smoking, and with quantitative estimates of exposure to outdoor air pollution. The follow-up of these studies extends from the 1970s to the 2010s.[43–46] During that period, outdoor air pollution levels generally declined in North America and Europe. Other, less informative studies included special populations, such as workers potentially exposed to air pollution in their jobs and studies based on ecological (group-level) definition of exposure, such as residence in a particular area.[47] Although case-control studies have the advantage of integrating exposure over the whole life of subjects, they might suffer from selection, recall, and information (exposure misclassification) biases to a larger degree than cohort studies, and the evidence derived from case-control studies is considered to be of lower quality compared with that from cohort studies.

Table 2. Results of Recent Meta-Analyses on Exposure to $PM_{2.5}$ and Risk of Lung Cancer

References	Study Design	Number of Studies Included	Outcome	RR (95% CI)
Hamra et al. (2014)[48]	Co, CC	14	IM	1.09 (1.04-1.14)
Cui et al. (2015)[49]	Co	12	IM	1.09 (1.06-1.11)
Yang et al. (2016)[50]	Co	10	IM	1.07 (1.01-1.13)
Chen and Hoek (2020)[51]	Co, CC	15	M	1.12 (1.07-1.16)
Ciabattini et al. (2021)[52]	Co	15	IM	1.16 (1.09-1.23)
Yu et al. (2021)[53]	Co	21	IM	1.16 (1.10-1.23)

Co, cohort; CC, case-control; CI, confidence interval; I, incidence; M, mortality; $PM_{2.5}$, particulate matter with aerodynamic diameter less than 2.5 microns; RR, relative risk for 10 $\mu g/m^3$ increase in $PM_{2.5}$ exposure.

Overall, cohort studies with quantitative measures of outdoor air pollution revealed positive associations with lung cancer in both sexes and from all geographic regions (Table 2[48–53]). The potential confounding effect exerted by tobacco smoking and other risk factors was accounted for by adjustment or stratification (Table 3[48,52,53]).

Most available studies estimated relative risks using multiplicative models with a continuous variable for $PM_{2.5}$ exposure, which assumes a linear exposure-response relationship. Evaluation of alternatives to a linear model was reported in some studies, which found no statistically significant deviation from linearity. For example, Turner et al.[6] found that each 10 $\mu g/m^3$ increase in $PM_{2.5}$ concentration was associated with a 15% to 27% increase in lung cancer mortality, even after adjusting for numerous variables, including smoking and secondhand smoke, occupational exposures (asbestos, coal dust, diesel engine exhaust, etc.), an occupational "dirtiness" index, and radon exposure, among others.

The results of these studies have been subject to numerous reviews and meta-analyses: results of recent meta-analyses are summarized in Table 2.[48–53] These meta-analyses differed with respect to inclusion criteria, but in general, they provided consistent results: the summary relative risk for a 10 $\mu g/m^3$ increase in $PM_{2.5}$ level (on the basis of the linear model discussed previously), ranged from 1.07 to 1.16, with more recent studies yielding higher risk estimates, which might reflect a decrease in exposure misclassification.

Some of the meta-analyses reported results are stratified by smoking status. These results are summarized in Table 3.[48,52,53] The summary relative risk was similar among those who never and ever smoked.

One limitation of some studies is the lack of adequate latency periods between exposure and outcome in analyses. Latency periods between air pollution exposure of disease diagnosis and mortality should differ. If the differences are ignored, biased overestimation of associations can occur.

A change in air pollution exposure takes 15 to 20 years to be reflected in the lung cancer rate as revealed by a study in Taiwan correlating the change in air pollution exposure between the northern and southern regions of the country and lung adenocarcinoma incidence rates (Fig. 1*A* and *B*).[54]

In both high-income countries and low-middle income countries (LMICs), socioeconomic factors were found to influence the risk of air pollution on lung cancer development. In the Pennsylvania Cancer Registry, the incidence of lung cancer was found to be increased in localities with higher population density, lower socioeconomic status, and greater proximity to major highways and manufacturing facilities.[55] In a nationwide study from the People's Republic of China, the association between $PM_{2.5}$ exposure and lung cancer incidence was stronger in urban areas and counties with lower economic and educational levels.[56] The greater risks associated with urban areas and proximity to highways suggest the vulnerability of traffic-exposed populations. In addition, a stronger association between PM exposure and lung cancer incidence has been observed in Chinese counties with higher temperatures and humidity.[56] This is important as many LMICs have a warm and humid climate.

Air pollution exposure after the diagnosis of lung cancer shortens lung cancer survival. When adjusted

Table 3. Results of Selected Meta-Analyses on Exposure to $PM_{2.5}$ and Risk of Lung Cancer, Stratified by Smoking Status

References	Never Smoked, RR (95% CI)	Ever Smoked, RR (95% CI)
Hamra et al. (2014)[48]	1.18 (1.00-1.39)	1.18[a] (0.89-1.57)
Ciabattini et al. (2021)[52]	1.15 (0.98-1.35)	1.24 (1.12-1.36)
Yu et al. (2021)[53]	1.17 (1.00-1.37)	1.21[a] (1.09-1.34)

[a]Based on meta-analysis of results for formerly smoked and currently smoking.
CI, confidence interval; $PM_{2.5}$, particulate matter with aerodynamic diameter less than 2.5 microns; RR, relative risk for 10 $\mu g/m^3$ increase in $PM_{2.5}$ exposure.

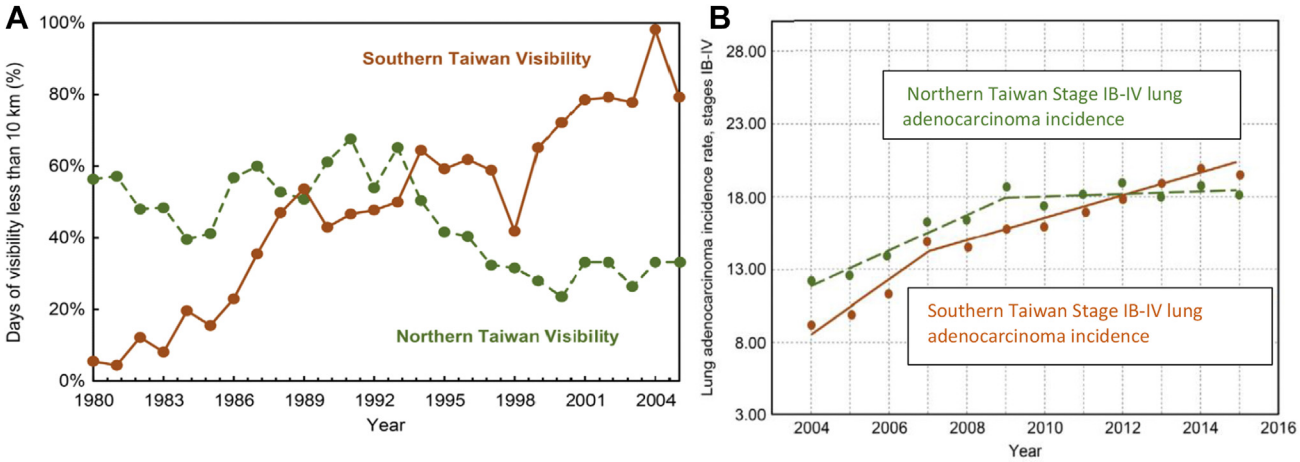

Figure 1. (*A*) Air pollution as estimated by days of visibility less than 10 km in Southern and Northern Taiwan, by year. (*B*) Incidence of lung adenocarcinoma stages IB to IV per 100,000 population (age adjusted) in Southern and Northern Taiwan, by year. Modified from reference.[54]

for demographic variables (age, sex, marital status, race/ethnicity), date of diagnosis, tumor characteristics at diagnosis (stage, anatomical site, histopathology), and treatment occurring less than 6 months after diagnosis, the median survival for patients with early stage lung cancer was 2.4 years for those with high $PM_{2.5}$ exposure ($\geq$16 $\mu g/m^3$) and 5.7 years for those with low $PM_{2.5}$ exposure ($<$10 $\mu g/m^3$).[57] The association of poor air quality and mortality in lung cancer has also been documented in another registry-based study in Pennsylvania, where exposure to higher levels of ambient air pollution (NO_2, $PM_{2.5}$, O_3, and PM_{10}) reduced survival in lung cancer.[58] This association seemed to be greatest for early stage tumors and adenocarcinomas.[57]

Although there is a decline in indoor air pollution globally, an estimated 2.5 billion people predominantly living in LMICs are still at risk due to solid fuel use for cooking and heating.[59] The IARC has designated coal as a group 1 and household biomass fuel as a group 2A human carcinogen.[60] Observational studies from the People's Republic of China and India have consistently revealed that individuals exposed to biomass fuel and coal have higher odds of developing lung cancer.[61–63] Furthermore, retrospective cohort studies from the People's Republic of China have found that transitioning from unvented coal stoves to either stoves with chimneys or portable stoves was associated with reductions in lifetime lung cancer incidence and mortality rates.[64,65] Fumes from cooking oil have also been found to increase the odds of developing lung cancer in a meta-analysis of 23 observational studies.[66]

Burden of Disease

In 2018, air pollution was responsible for approximately 8.7 million deaths worldwide—one in five deaths globally.[67] It has been estimated that 265,267 lung cancer deaths worldwide were attributed to $PM_{2.5}$ pollution, accounting for 14.1% of all lung cancer deaths in 2017.[68]

Recently, the WHO lowered the annual mean air quality guideline for $PM_{2.5}$ from 10 $\mu g/m^3$ to 5 $\mu g/m^3$.[69] Around the world, the average annual concentrations of $PM_{2.5}$ currently range from less than 10 $\mu g/m^3$ in countries such as Canada to more than 100 $\mu g/m^3$ in South and East Asia with a global population-weighted annual average of 46 $\mu g/m^3$.[70,71] An Australian study revealed that long-term (16 y) exposure to $PM_{2.5}$ increases respiratory and other cause-specific mortality especially in females at a level as low as 4.5 $\mu g/m^3$.[72] Although the average general residential levels of $PM_{2.5}$ in countries such as Canada are less than 10 $\mu g/m^3$, associations between low levels of $PM_{2.5}$ exposure and increased Canadian overall mortality have been reported.[73] A recent study of more than 7.1 million Canadians reported a significant association between overall mortality and chronic exposure to doses of $PM_{2.5}$ below current standards and guidelines, including the most stringent WHO air quality guideline of 5 $\mu g/m^3$.[74] Exposure to air pollutants early in life has long-lasting detrimental effects. For example, exposure to traffic air pollutants during infancy can have an impact on lung function up to adolescence.[75] With growing and urbanizing populations, a warmer climate, and increasingly frequent wildfires,[76] many cities in the world will exceed the 5 $\mu g/m^3$ threshold, resulting in air pollution becoming an even larger public health problem and an increased lung cancer burden. Air pollution disproportionately affects people of lower socioeconomic status. In a study in the United States, the burden of lung cancer and other causes of death associated with $PM_{2.5}$ was found to be

disproportionally borne by socioeconomically disadvantaged communities and black individuals.[77]

Association Between Climate Change and Air Pollution

Air pollution and climate change are not separate issues. There are close interactions and synergistic effects of the two together on population health and lung cancer. It is well established that burning fossil fuels with the subsequent release of greenhouse gases causes both climate change and air pollution. Greenhouse gases such as methane, carbon dioxide, and nitrous oxides trap infrared radiation and radiate it back to the Earth's surface, warming the planet. Air pollution causes chain effects with detrimental impacts on multiple environmental aspects, including surrounding air, soil, and water, in both urban and rural areas.[78] Air pollution is worse when superimposed with certain climatological conditions including strong sunlight (ultraviolet light) and light wind. This facilitates the formation and accumulation of photochemical smog, resulting from photochemical reaction of ultraviolet acting on nitric oxides and VOCs. Ground-level O_3 is usually formed as a byproduct of photochemical smog.

$PM_{2.5}$ can also come from events worsened by climate change, such as wildfires and sandstorms. $PM_{2.5}$ from wildfires is up to 10 times more harmful on human health than $PM_{2.5}$ from other sources because it is mostly carbonaceous (with 5%–20% elemental carbon and at least 50% organic carbon) and has more oxidative potential than ambient urban particulate due to the presence of more polar organic compounds.[79] Most large fires in the United States occur in the West, but one study found that most mortality (74%) and asthma morbidity (approximately 75%) attributable to smoke $PM_{2.5}$ occur outside the West in heavily populated areas east of the Mississippi.[80] Wildfire exposure has also been associated with increased incidence of lung cancer. One population-based observational cohort study in Canada found that people exposed to a wildfire within 50 km of residential locations in the past 10 years had a 4.9% relative higher incidence (hazard ratio: 1.049; 95% confidence interval: 1.028–1.071) of lung cancer than the unexposed population.[81]

Primary Prevention: Reducing Human Exposure to Air Pollution

As awareness of the effects of air pollution on lung health increases, establishing national, regional, and local public health guidelines on the basis of scientific evidence has become an imperative. Although the most beneficial solution would be to reduce air pollution at the source, this is not always feasible. In 2019, the WHO held an expert consultation on the topic of personal interventions and risk communication, addressing mainly the communication of scientific data, air pollution levels, and the health impact thereof, including individual responsibilities, also stating that suggested measures should address all questions of equity.[82]

Carlsten et al.[83] reviewed available literature and proposed several evidence-based suggestions, albeit none of the recommendations were above a level C strength. Nevertheless, the strategies offered were simple measures that could be recommended to the population, or to those who have risk factors. These included being aware of air pollution $PM_{2.5}$ or PM_{10} measurements which can be found at the following website: https://aqicn.org/map/world/.

In addition, useful measures include wearing a close-fitting particulate respirator if air pollution levels are high, shifting to nonmotorized alternative transportation such as cycling and walking, choosing a route that minimizes near-road pollution when possible, and not opening windows of vehicles in traffic. To minimize household pollution, switching to clean fuels and maintaining adequate ventilation were suggested, including using air purifiers.[83]

A key issue is communicating the health hazards. Although both European and U.S.–based environmental protection agencies[84,85] and many others publish well-visualized data on current and forecasted air pollution in many regions or cities with guidance on potential interventions for the public, for most of the world, these resources remain underused.

Laumbach and Cromar[86] argued that not only the level of air pollution should be considered while offering public guidance, but also its frequency. Interventional measures should be based on whether the elevated levels of particles occur only episodically or are a regular occurrence in the region. Air pollution is a complex phenomenon, with various causes and a wide array of pollutants resulting in diverse mixtures. Personal susceptibility also varies greatly in as-yet poorly understood manners. Therefore, any guidance given—especially when addressing a larger population—should account for this uncertainty, weighing not only the benefits but also harmful consequences.[86]

Incorporation of Air Pollution Into Lung Cancer Risk Prediction Models

Lung cancer screening with low-dose computed tomography (LDCT) has been found to reduce lung cancer mortality by 20% or more in high-risk individuals.[87,88] LDCT lung screening may be less effective in low-risk individuals.[89,90] For the most efficient lung cancer screening, it is important to identify individuals truly at high risk. Lung cancer risk prediction models are increasingly playing important roles in lung cancer screening, by identifying individuals at high risk who are

most likely to benefit from LDCT lung screening and for estimating risks to facilitate informed risks/harms-benefits discussions between potential screening participants and care providers.

Air pollution, in particular $PM_{2.5}$, is now regarded as an established cause of lung cancer.[7,8] Global, outdoor (ambient) air pollution is regarded as the second most important cause of lung cancer mortality after smoking exposure, and indoor air pollution is considered the seventh most important cause.[68] Neither outdoor nor indoor air pollution is included as predictor in any validated lung cancer risk prediction models to date. Ambient exposure to $PM_{2.5}$ may be a particularly useful additional predictor in new or existing lung cancer risk prediction models. Given the synergistic effect of air pollution and lung cancer, those ever-smoker individuals below current risk thresholds may particularly benefit. Ten Haaf and de Koning[91] presented a microsimulation modeling analysis suggesting that individuals who never smoked can potentially benefit from screening similar to individuals with a smoking history if they are identified at high risk. Including air pollution in future risk prediction models may help identify individuals who have never smoked who have high enough risk to benefit from screening.

Several issues exist with the inclusion of air pollution in lung cancer risk models.

Estimation of cumulative exposure to air pollution exposure is complex but is readily conducted for research purposes and for studying associations. Nevertheless, to be useful in prediction models intended for use in clinical and public health settings to estimate individuals' risks, models need to be parsimonious and predictor data need to be collected simply and quickly. This poses a major challenge with accurately collecting long-term exposure to ambient air pollution and simply incorporating it in prediction models. Another challenge in incorporating air pollution into risk prediction models is that the effect and impact of air pollution in different regions of the world may vary, and different models may be required in different regions.

Globally, the impact of indoor air pollution is smaller than outdoor air pollution, but this may not be the case in certain subsets of a population, for example, in East Asian never-smoking women. Assessing risk in these subgroups may require a tailored model with multiple air pollution exposure measurements.

Currently, modeling studies are underway in an attempt to address these and other challenges.

Role of Advocacy by Health Professionals

Health care professionals are trusted and can make a difference in describing the health impacts of air pollution and climate change and advocating for reducing the amount of fossil fuel burned which affects both air pollution and greenhouse gases. They can make these effects more personal by translating air pollution effects into health consequences at the local level and by telling stories about the impact on individuals. In conjunction with health care economists, they can help to monetize the cost of increased morbidity and mortality (e.g., premature death, hospital room visits, increase in chronic conditions) due to air pollution and global warming.[92] This allows use of cost-effectiveness models to guide expenditures for amelioration and mitigation. A multinational survey of health care professionals found that 75% of participants said that the issue of climate change is very important or extremely important to them, whereas 86% somewhat or strongly agreed that health professionals have a responsibility to explain the effects of climate change on health to the general public, and 89% agreed that health professionals should encourage their national leaders and global leaders to strengthen commitments to the Paris Agreement. Nevertheless, they reported multiple barriers to engaging in advocacy, including limited time (most cited barrier), lack of knowledge, and lack of support or belief that their efforts would not make a difference. Things they said would be helpful included continuing professional education, policy statements by their professional organizations, and guidance on how to reduce the carbon footprint in their workplace.

Medical societies can play a key role in reducing the barriers for climate advocacy for health professionals and in influencing legislation on air pollution and climate change by joining together and creating a coalition to develop position and policy statements about air pollution and work with local legislatures to sponsor changes to climate policy. Medical providers depend on these societies to provide direction in health care policy and best practices.[93] A review of medical society public facing websites revealed that only 22 of the 111 websites (11%) reviewed included a position or policy statement about climate change. The Medical Society Consortium on Climate and Health is a U.S. organization focused on engaging medical societies to raise awareness among their members and advocate for enhanced environmental policies and decreased fossil fuel emissions.[94] In addition, in the United States, the American Lung Association urges medical professionals to join the health professionals for clean air and climate action campaign.[95] Similarly, in the United Kingdom, doctors are being asked to become advocates resulting in the creation of the first U.K. environmental air pollution clinic for children.[96] The call to action includes five things general practitioners can do to reduce air pollution.[96]

In addition, health care is the second largest user of energy after the food industry. Individual institutions can make changes to reduce their carbon footprint. The Agency for Healthcare Research and Quality has developed a primer to help institutions reduce their greenhouse gas emissions in six domains. These include building energy, transportation, anesthetic gas, pharmaceuticals and chemicals, medical devices and supplies, and food.[97] The primer includes case studies of what organizations have done to reduce their greenhouse gas emissions. One particular example for respiratory physicians is the benefits of switching from metered-dose inhalers to alternative low global-warming potential inhalers that is also accompanied by cost-savings.[98]

Globally, the WHO has created the WHO Alliance for Transformative Action on Climate and Health (ATACH).[99] ATACH was created to "…realize the ambition set at COP26 to build climate resilient and sustainable health systems, using the collective power of WHO Member States ("Member States") and other stakeholders to drive this agenda forward at pace and scale; and promote the integration of climate change and health nexus into respective national, regional, and global plans." Individual organizations with expertise in climate change are encouraged to apply to join ATACH.

Other medical societies focused on climate change include *Healthcare Without Harm*, a global organization focused on transforming health care to reduce its environmental footprint.[100] Health professionals are also able to join the *My Green Doctor Foundation*, a not-for-profit organization created in 2021 "…to help educate health professionals and patients, and to motivate them to make changes in their lives, on topics of environmental sustainability and climate change." The foundation provides educational tools to health professionals and patients to guide actions on climate change.[101]

Next Steps and Conclusions

The impacts of air pollution observe no geographic boundaries. Air pollution causes many health hazards and can contribute to the development of lung cancer and worsen its prognosis. Awareness that exposure to air pollution is the second largest risk factor for lung cancer, second only to tobacco, needs to be at the forefront of the lung cancer community to recognize its risks and to educate patients and the health care community at large.

Sources of air pollution and climate change are often the same. Sectors responsible include energy production, industry, transportation, agriculture, cooking, and heating. The health care sector alone is estimated to contribute to 8% of worldwide greenhouse gas emissions. Worryingly, the natural disasters from climate change can also adversely affect the well-being of people with cancer and their survival.

Close international collaborative efforts in research and advocacy are urgently needed to lower the toll of air pollution and climate change, which will also benefit patients with lung cancer, their families, and the rest of the society. The International Association for the Study of Lung Cancer (IASLC) issued a strong, clear statement on air pollution and lung cancer.[102] IASLC calls for air pollution targets to be at the lowest levels as recommended by the World Health Organization. IASLC also calls on the health care community to practice environmental sustainability with low carbon emissions. Continued research and collaboration from everyone in the health care community is needed to realize the health benefits of cleaner skies.

CRediT Authorship Contribution Statement

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